

KNOX COUNTY REGIONAL, ME, USA

WMO: 726079

Lat: 44.067N

Lon: 69.100W

Elev: 14

StdP: 101.15

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94601

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-15.9	-27.1	0.3	-17.7	-23.7	0.4	-14.1	12.1	-1.5	11.1	-2.9	3.3	290	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	27.6	20.8	26.1	19.7	24.1	18.7	22.2	26.1	21.0	24.4	19.9	23.0	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	15.7	24.6	19.6	14.3	23.0	18.7	13.6	22.0	64.9	26.1	60.5	24.4	57.0	23.2	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.5	7.5	-21.4	31.3	3.2	1.8	-23.7	32.6	-25.5	33.6	-27.3	34.6	-29.6	35.9
			-22.0	24.1	3.2	1.2	-24.4	25.0	-26.3	25.7	-28.1	26.4	-30.4	27.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-5.0	-3.8	0.1	5.5	10.6	15.3	18.9	18.5	14.8	9.2	3.9	-1.4
	DBStd	9.19	6.20	5.32	4.59	3.35	3.25	3.27	2.66	2.54	3.40	3.72	4.33	5.02
	HDD10.0	1937	466	385	309	140	29	2	0	0	3	61	188	355
	HDD18.3	4134	724	618	566	386	240	100	25	29	114	284	434	613
	CDD10.0	943	0	0	1	5	49	162	276	262	148	36	4	1
	CDD18.3	96	0	0	0	0	2	11	42	33	8	0	0	0
	CDH23.3	575	0	0	1	1	22	83	253	165	47	3	0	0
	CDH26.7	99	0	0	0	0	4	15	49	23	7	0	0	0
(14) Wind	WSAvg	3.3	3.8	3.7	3.8	3.7	3.3	2.8	2.5	2.4	2.7	3.3	3.5	3.6
(15) Precipitation	PrecAvg	1326	112	98	118	117	102	100	89	78	118	131	132	129
	PrecMax	1876	207	213	241	261	240	242	190	196	287	405	232	220
	PrecMin	870	16	50	27	11	21	45	38	13	44	26	65	47
	PrecStd	249	49	34	52	62	49	54	41	47	68	81	40	41
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	9.8	15.3	18.8	26.5	28.7	31.0	29.0	27.4	22.2	16.0	12.5
		MCWB	10.1	7.7	9.6	11.4	17.9	21.3	23.2	23.0	20.2	16.1	12.9	10.6
	2%	DB	8.5	7.5	10.6	15.1	21.5	26.1	27.7	27.0	24.1	18.3	13.9	10.2
		MCWB	7.8	5.4	6.3	8.6	15.4	18.2	21.1	20.6	18.8	14.4	11.6	9.0
	5%	DB	6.5	6.0	7.8	12.7	18.6	23.0	26.2	25.2	22.3	17.0	12.5	8.0
		MCWB	4.8	3.9	5.0	7.9	13.2	17.1	20.1	19.4	17.7	13.9	10.6	6.5
	10%	DB	3.8	3.9	6.3	11.2	16.4	21.4	24.1	23.6	20.6	15.2	11.2	6.4
		MCWB	2.1	2.0	3.6	7.2	11.4	16.1	19.0	18.6	16.8	12.6	9.5	4.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	8.3	10.5	12.9	19.0	22.3	24.0	23.7	22.2	17.6	14.1	11.4
		MCDB	10.5	9.0	14.2	17.1	24.3	27.4	28.8	27.8	26.2	19.5	15.0	12.0
	2%	WB	7.8	6.1	7.6	10.6	16.3	20.2	22.3	21.9	20.1	16.0	12.4	9.0
		MCDB	8.2	7.3	8.9	12.8	20.3	24.5	26.5	25.1	22.7	17.6	13.3	9.9
	5%	WB	5.1	4.2	5.7	8.9	14.1	18.5	21.1	20.8	18.8	14.6	11.1	7.3
		MCDB	6.6	5.7	7.7	11.5	17.7	21.9	24.8	23.6	21.0	16.5	12.1	8.0
	10%	WB	2.5	2.6	4.1	7.8	12.5	17.1	20.0	19.7	17.7	13.1	9.4	4.9
		MCDB	3.7	3.9	6.1	10.1	15.2	19.8	23.2	22.4	19.2	15.0	11.0	6.6
(35) Mean Daily Temperature Range	MDBR		8.7	8.9	8.0	8.2	8.4	8.6	8.9	9.1	9.2	8.7	7.9	8.3
	5% DB	MCDBR	9.8	9.3	9.6	11.4	12.8	12.0	11.5	11.1	11.3	10.1	8.8	10.2
		MCWBR	10.0	8.3	7.2	7.2	7.6	7.0	6.3	6.5	7.1	7.3	7.8	9.8
	5% WB	MCDBR	9.9	9.0	8.9	9.2	11.0	10.7	9.9	9.6	9.2	8.8	7.9	10.0
		MCWBR	10.3	8.6	7.4	6.9	7.4	7.2	6.2	6.2	6.4	7.3	7.8	10.4
(40) Clear-Sky Solar Irradiance	taub		0.297	0.305	0.328	0.354	0.380	0.422	0.441	0.410	0.367	0.338	0.319	0.298
	taud		2.469	2.457	2.461	2.425	2.396	2.291	2.260	2.336	2.437	2.518	2.535	2.515
	Ebn at Noon		835	892	913	911	891	851	829	844	858	843	803	800
	Edh at Noon		71	85	97	109	116	129	132	118	99	80	67	62
(44) All-Sky Solar Radiation	RadAvg		1.60	2.49	3.69	4.61	5.23	5.63	5.90	5.32	4.28	2.75	1.71	1.26
	RadStd		0.12	0.20	0.23	0.49	0.63	0.47	0.43	0.28	0.26	0.25	0.15	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+63	N/A

Nomenclature: See separate page